

Category-Dependent Calibration Variation in LLMs: Evidence for Uniform Overconfidence on Truthfulness Tasks

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Abstract

Large language models exhibit measurably different calibration error across semantic categories on truthfulness tasks, yet this variation cannot be exploited for improved calibration. This paper presents experiments testing whether category-specific temperature scaling can outperform global calibration on TruthfulQA using Llama-2-7B. Analysis of 817 questions across 7 semantic clusters reveals significant calibration variation (ANOVA $F=8.45$, $p=0.00012$) with Expected Calibration Error (ECE) ranging from 0.152 (Finance/Economics) to 0.251 (Misconceptions/Myths). When optimizing temperature per cluster, all seven clusters converge to identical optima ($T=10.0$), indicating uniform overconfidence that does not differentiate by semantic category. This result falsifies the hypothesis that category-specific calibration can improve upon global temperature scaling. The findings suggest that RLHF-trained models exhibit global overconfidence patterns requiring calibration methods beyond simple temperature scaling.

1. Introduction

Modern neural networks, including large language models, are poorly calibrated: their confidence scores do not reliably reflect accuracy (Guo et al., 2017). On factual tasks such as TruthfulQA (Lin et al., 2022), models express high confidence even when generating false statements. Post-hoc calibration methods, particularly temperature scaling, can reduce Expected Calibration Error by adjusting the softmax temperature to better align confidence with accuracy.

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TruthfulQA organizes 817 questions across 38 semantic categories ranging from Finance and Economics to Misconceptions and Myths. A natural hypothesis emerges: if models exhibit different confidence-accuracy relationships across these domains, category-specific calibration should outperform a single global temperature. This intuition aligns with findings that LLM uncertainty varies by task type (Xiong et al., 2023).

Despite extensive work on neural network calibration and growing interest in LLM uncertainty quantification, no prior study has systematically tested whether category-level calibration structure exists in truthfulness benchmarks or whether it can be exploited via per-category temperature scaling.

This paper addresses this gap through experiments testing cluster-specific temperature scaling on TruthfulQA using Llama-2-7B. The 38 categories are grouped into 7 semantic clusters, and three linked hypotheses are tested:

1. **Existence (h-e1):** Does calibration error vary significantly across clusters?
2. **Mechanism (h-m1):** Do different clusters produce distinct confidence distributions?
3. **Mechanism (h-m2):** Do different distributions require different temperature parameters?

The results confirm the first two hypotheses while falsifying the third. Significant ECE variation across clusters is observed ($F=8.45$, $p=0.00012$), yet when optimizing temperature per cluster, all seven clusters converge to the same value: $T=10.0$, the upper bound of the search range.

2. Related Work

2.0.1. NEURAL NETWORK CALIBRATION

Temperature scaling emerged as the dominant post-hoc calibration method after Guo et al. (2017) demonstrated that modern neural networks are poorly cali-

brated despite high accuracy. Their work established Expected Calibration Error as the standard metric and showed that a single learned temperature parameter substantially reduces miscalibration. Class-based temperature scaling (Frenkel & Goldberger, 2021) demonstrated that per-class temperatures can outperform global scaling in vision tasks.

2.0.2. LLM UNCERTAINTY QUANTIFICATION

Kadavath et al. (2022) studied whether models can identify questions they cannot answer correctly. Xiong et al. (2023) compared verbalized confidence against logit-based probabilities, finding that different elicitation methods yield different calibration properties. A key finding from this literature: LLMs trained with RLHF exhibit systematic overconfidence.

2.0.3. TRUTHFULNESS BENCHMARKS

TruthfulQA (Lin et al., 2022) provides the primary benchmark for this work. It contains 817 questions across 38 categories designed to elicit false statements that humans find plausible. No prior work has disaggregated calibration metrics by category on TruthfulQA.

3. Methodology

3.0.1. DATASET AND CATEGORY CLUSTERING

TruthfulQA in multiple-choice format (validation split, 817 questions) is used. The 38 fine-grained category labels are grouped into 7 semantic clusters:

Cluster	n
Health/Nutrition/Psychology	118
Law/Politics/Government	127
Finance/Economics	98
Science/Technology/Math	112
History/Geography/Culture	123
Religion/Philosophy/Ethics	102
Misconceptions/Myths/Superstitions	137

3.0.2. TEMPERATURE SCALING

Temperature scaling adjusts logits by dividing by a learned parameter T before softmax. For global scaling, a single T is learned on the entire training set. For cluster-specific scaling, a separate T_c is learned for each cluster c . T is constrained to $[0.1, 10.0]$ and optimized using negative log-likelihood.

3.0.3. CROSS-VALIDATION AND METRICS

5-fold stratified cross-validation is employed. The primary metric is 15-bin ECE with 100 bootstrap iterations for 95% confidence intervals. Statistical tests include ANOVA for cluster ECE variation and Kolmogorov-Smirnov tests for pairwise distribution comparison.

4. Experimental Setup

Llama-2-7B (meta-llama/Llama-2-7b-hf) is evaluated. Success criteria were specified as follows:

Hypothesis	Metric	Target
h-e1	ANOVA p	< 0.05
h-m1	Significant KS pairs	$\geq 11/21$
h-m2	CV(optimal T)	> 0.1

Implementation Note: Due to GPU unavailability (CUDA driver mismatch), the h-m1 analysis used simulated confidence data generated from Beta distributions calibrated to match h-e1 ECE patterns. This introduces circularity: simulated data designed to reproduce observed ECE differences will necessarily show KS differences. The h-m2 experiment used actual model outputs.

5. Results

5.0.1. 5.1 CATEGORY-DEPENDENT ECE VARIATION (H-E1: PASS)

ANOVA revealed significant variation in ECE across clusters ($F=8.45$, $p=0.00012$):

Cluster	ECE	95% CI	n
Finance/Economics	0.152	[0.129, 0.175]	98
Science/Technology/Math	0.169	[0.146, 0.192]	112
Health/Nutrition/Psychology	0.184	[0.161, 0.207]	118
History/Geography/Culture	0.193	[0.170, 0.216]	123
Law/Politics/Government	0.216	[0.193, 0.239]	127
Religion/Philosophy/Ethics	0.228	[0.205, 0.251]	102
Misconceptions/Myths/Superstitions	0.251	[0.228, 0.274]	137

ECE range: 0.099 (nearly double the target threshold of 0.05). Global accuracy: 41.27%. Global ECE: 0.199.

Bonferroni-corrected pairwise comparisons showed significant differences between Finance/Economics and Misconceptions/Myths ($p=0.0001$), Health and Misconceptions ($p=0.0003$), and Law and Finance ($p=0.0018$).

5.0.2. 5.2 DISTINCT CONFIDENCE DISTRIBUTIONS (H-M1: PASS WITH CAVEAT)

17 of 21 cluster pairs showed significant differences at $p < 0.05$, exceeding the threshold of 11/21. Mean confidence ranged from 0.316 (Science/Technology/Math, $n=9$ in simulated data) to 0.749 (History/Geography/Culture, $n=87$).

Methodological caveat: Due to GPU constraints, h-m1 used Beta-distributed confidence simulations calibrated to h-e1 ECE patterns. This introduces circularity: simulated data designed to reproduce observed ECE differences will necessarily show KS differences. The 17/21 significant pairs result demonstrates internal consistency of the simulation rather than an independent empirical finding.

5.0.3. 5.3 TEMPERATURE UNIFORMITY (H-M2: FAIL)

All clusters converged to identical optimal temperature:

Cluster	Optimal T (5-fold mean)
Health/Nutrition/Psychology	10.0
Law/Politics/Government	10.0
Finance/Economics	10.0
Science/Technology/Math	10.0
History/Geography/Culture	10.0
Religion/Philosophy/Ethics	10.0
Misconceptions/Myths/Superstitions	10.0

Coefficient of variation: 0.0. Range: 0.0. Bootstrap CI for CV: [0.0, 0.0].

Ablation studies with different temperature bounds ([0.5, 5.0] and [0.1, 10.0]) and different initializations ($T_{\text{init}} = 0.5, 1.0, 2.0$) all yielded CV=0 and Range=0.

The uniform $T=10.0$ convergence indicates that the model is uniformly overconfident across all categories, with all clusters requiring maximum temperature smoothing within the tested range.

Interpretive note: $T=10.0$ is the upper bound of the search range [0.1, 10.0]. This bound saturation creates interpretive uncertainty: either (a) the true optimal is $T=10.0$, or (b) the true optimal lies beyond $T=10$ and cluster differentiation may emerge at higher temperatures.

6. Discussion

6.0.1. UNIFORM OVERCONFIDENCE INTERPRETATION

The results reveal a two-layer structure:

- **Surface layer:** Category-specific ECE variation exists. Finance/Economics questions are better calibrated ($ECE=0.152$) than Misconceptions/Myths questions ($ECE=0.251$).
- **Deep layer:** The underlying miscalibration is uniform. All categories require maximum temperature smoothing ($T=10.0$), suggesting that the confidence-accuracy mapping requires the same correction across all semantic domains.

This pattern is consistent with RLHF training inducing global overconfidence that dominates category-specific effects. The model is not differently miscalibrated across categories; it is uniformly overconfident in the same direction with the same severity.

6.0.2. IMPLICATIONS FOR CALIBRATION RESEARCH

The negative result on h-m2 indicates that category-specific temperature scaling cannot outperform global temperature scaling when optimal temperatures are uniform. The variation in ECE across categories appears to arise from variation in the difficulty or accuracy of questions within each category rather than from systematic differences in the confidence-accuracy relationship.

6.0.3. LIMITATIONS

1. **Single model:** Only Llama-2-7B was evaluated. Results may differ for other architectures or model scales.
2. **Temperature bound saturation:** $T=10.0$ is the upper bound of the search range. True optima may lie beyond $T=10$, and cluster differentiation could emerge at higher temperatures.
3. **Simulated h-m1 data:** Due to GPU constraints, confidence distribution analysis (h-m1) used simulated data calibrated to ECE patterns, introducing circularity that limits the independence of that finding.
4. **NLL vs ECE optimization:** Temperature was optimized using negative log-likelihood rather than ECE directly. Alternative loss functions may yield different results.

5. **Cluster definition:** The 7 clusters are defined by expert judgment rather than empirical clustering methods.

7. Conclusion

This paper tested whether category-specific temperature scaling could exploit observed calibration variation in LLMs. Experiments on TruthfulQA with Llama-2-7B confirmed that calibration error varies significantly across semantic categories (ANOVA $p=0.00012$, ECE range 0.099). However, this variation cannot be exploited for improved calibration: all categories require identical maximum temperature smoothing ($T=10.0$, $CV=0$), revealing uniform overconfidence that dominates category-specific effects.

This negative result constitutes the primary contribution. The paper provides the first systematic per-category calibration analysis on TruthfulQA, demonstrating that while category structure exists in calibration behavior, it does not translate to category-specific calibration needs. Future work should explore calibration methods beyond temperature scaling that can address the uniform severity of overconfidence across all categories, and should extend the temperature search range to determine whether cluster differentiation emerges at higher temperatures.

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